FUNDAMENTALS OF DATA SCIENCE CASE STUDY REPORT

Case Study 1 and Case Study 4

Case Study 1: Online Retail Baskets · Case Study 4: Student Dropout and Academic Success

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Abstract

This report documents two analyses carried out on the datasets supplied for the course. **Case Study 1** examines 7,501 online-retail baskets. It profiles the products and basket sizes, measures how strongly demand is concentrated, mines the most common product pairs, encodes every basket as a binary vector, reduces it with PCA and compares four clustering algorithms (K-means, hierarchical, DBSCAN and spectral) with the silhouette coefficient and the Davies-Bouldin index. **Case Study 4** examines 4,424 student records labelled Dropout, Enrolled or Graduate. It audits the data, compares admission grade, first-semester grade and age across outcomes, tests the first-semester difference with the Kruskal-Wallis test followed by Bonferroni-adjusted pairwise tests, and evaluates a multiclass logistic-regression model that uses only information available after the first semester.

Main findings: (i) demand is concentrated, as 17 of 119 products account for half of all product appearances, and the typical basket is small (median 3 items, 23.4% single-item); (ii) no clustering method separates baskets cleanly, because every silhouette value stays below 0.09; (iii) first-semester grade differs strongly across outcomes ($H = 1{,}094.19$, $p < 0.05$, effect size about 0.25), with 40.1% of dropouts holding a first-semester grade of zero against 3.5% of graduates; and (iv) the baseline classifier reaches 74.5% accuracy and macro-F1 0.668 on held-out data but finds only 35% of Enrolled students, a gap that class re-weighting only partly closes.

Contents

Part	Topic	What it contains
Preliminary	Scope and workflow	Report structure and the data-to-insight process followed
Case Study 1	Online retail baskets	Data audit, basket statistics, product frequency and concentration, product pairs, PCA, four clustering algorithms, sensitivity checks, implications, limitations
Case Study 4	Student dropout and academic success	Data audit, variable families, exploratory analysis, hypothesis tests, classification model, per-class evaluation, coefficients, recommendations
Closing	Discussion	Cross-case conclusions and reproducibility checklist

Analytical workflow

Stage	Guiding question	Methods applied
Inspect	What do the rows and fields represent?	Shape, header check, data types, missing values, duplicates, unusual values
Prepare	How should the data be represented?	Basket lists and multi-hot vectors; one-hot encoding of codes; scaling inside a pipeline
Explore	What patterns appear?	Counts, distributions, concentration curve, pair co-occurrence, grouped summaries, box and violin plots
Evaluate	How reliable are the patterns and predictions?	Silhouette, Davies-Bouldin, repeated-sample stability, Kruskal-Wallis, pairwise tests, held-out and cross-validated metrics
Interpret	What can be concluded responsibly?	Effect sizes, model coefficients, limitations, association versus causation

CASE STUDY 1 - ONLINE RETAIL BASKETS

1.1 Problem statement

The retailer stores each transaction as a list of products bought together, but it has no structured picture of what a typical basket contains or whether baskets fall into recognisable groups. Without that picture, stocking, promotion and shelf-layout decisions are made by habit. This study therefore asks three questions: which products dominate, how large baskets are, and whether baskets form distinct groups once they are expressed numerically.

1.2 Objectives

- Understand and clean the retail file with Pandas.
- Compute descriptive statistics for basket size and identify the most frequent products.
- Visualise product frequencies and the basket-size distribution.
- Represent each basket as a numeric vector and reduce it with PCA.
- Apply K-means, hierarchical, DBSCAN and spectral clustering and compare them with the silhouette coefficient and the Davies-Bouldin index.
- Discuss limitations and draw cautious business implications.

1.3 Dataset description

The CSV file has 7,501 lines and 20 comma-separated fields per line. A line is one transaction and its fields hold the products bought, left to right; shorter baskets leave the remaining fields blank. **The file has no header row.** The first line (shrimp, almonds, avocado and so on) is a genuine 20-item basket, so the file was read with header=None. Reading it with the default header setting would silently discard that basket and shift every count by one transaction.

Attribute	Observed value	Comment
Transactions (rows)	7,501	One basket per row; no customer identifier
Product fields per row	20	Upper limit on basket size
Non-empty cells	29,363	Product entries before removing repeats
Empty cells	120,657	80.4% of 150,020 cells, so the data are sparse
Repeated product inside one basket	5	Removed; a basket is treated as a set of products
Distinct products	119	After trimming spaces and lower-casing labels
Baskets identical to an earlier basket	2,347 (31.3%)	Expected, since small baskets of popular items repeat; kept, because repeats reflect real demand

1.4 Data preparation

1. **Read** the file without a header and as text, so that no product name is mistaken for a column label.
2. **Clean** every cell: strip whitespace, convert to lower case and treat blank cells as missing. One product label carried stray spacing, which would otherwise split a product into two vocabulary entries.
3. **Build baskets:** collect the non-missing cells of each row into a list and drop repeated products within the list.
4. **Count** appearances per product and items per basket.
5. **Vectorise:** create one 0/1 column per product, giving a 7,501 x 119 matrix in which only 3.3% of the entries are 1.

The binary encoding compares baskets by which products they share and avoids implying any numeric order among product names. Its cost is that quantities are lost, because the file does not record them. The unit of analysis is the basket, not a person.

1.5 Descriptive statistics of basket size

Statistic	Value	Reading
Mean	3.91 items	Average basket size
Median	3 items	Half of all baskets have this many items or fewer
Mode	1 item	Single-item baskets are the most common size (1,754 baskets, 23.4%)
Quartiles (Q1 / Q3)	2 / 5 items	The middle half of baskets holds 2 to 5 items
Minimum / Maximum	1 / 20 items	The largest basket fills all 20 available fields
Standard deviation	2.90 items	Spread around the mean
Skewness	1.19	Positive, so the right tail is long

The mean (3.91) exceeds the median (3) and the mode is 1, the signature of a right-skewed distribution: many small baskets and a few large ones that pull the average up. For planning the median and quartiles describe a normal visit better than the mean. 55.4% of baskets contain three items or fewer, while only 256 baskets (3.4%) exceed ten items.

1.6 Product frequency and concentration

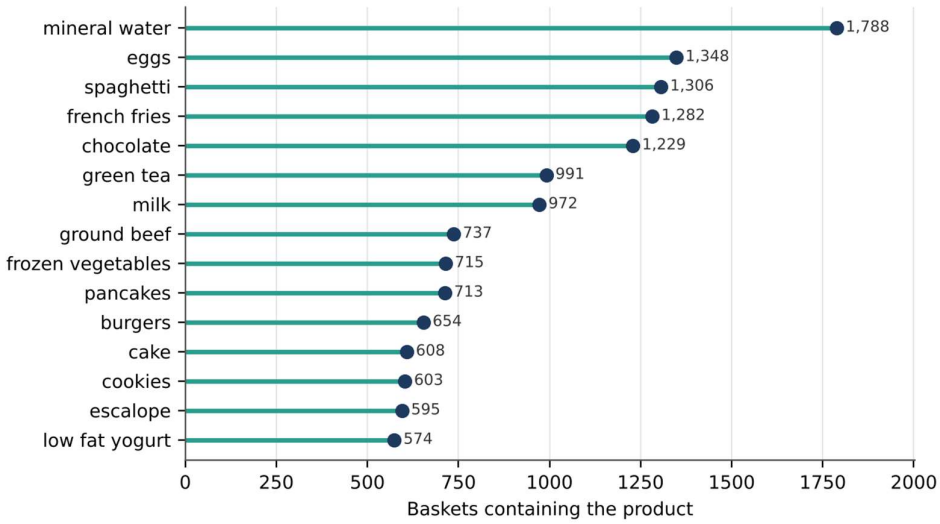


Figure 1. The fifteen products that appear in the most baskets (labels show basket counts).

Mineral water appears in 1,788 baskets (23.8%), well ahead of eggs, spaghetti, french fries and chocolate, which each appear in roughly 1,200 to 1,350 baskets (16% to 18%). Below the top five the counts fall away quickly: green tea and milk appear in about 970 to 990 baskets, and from ground beef downward every product appears in fewer than 740.

Rank	Product	Baskets	Share of baskets
1	mineral water	1,788	23.8%
2	eggs	1,348	18.0%
3	spaghetti	1,306	17.4%
4	french fries	1,282	17.1%
5	chocolate	1,229	16.4%
6	green tea	991	13.2%
7	milk	972	13.0%
8	ground beef	737	9.8%
9	frozen vegetables	715	9.5%
10	pancakes	713	9.5%

Demand is strongly concentrated (right panel of Figure 2). The five leading products account for 6,953 of 29,358 product appearances (23.7%); **17 products cover half of all appearances and 45 cover 80%**, leaving roughly 74 products to share the last fifth. The rarest items (tea, bramble, cream, napkins and water spray) each appear in 29 or fewer baskets. Frequency says how often something is bought, not how profitable it is or whether it is bought because of another item.

1.7 Basket-size distribution

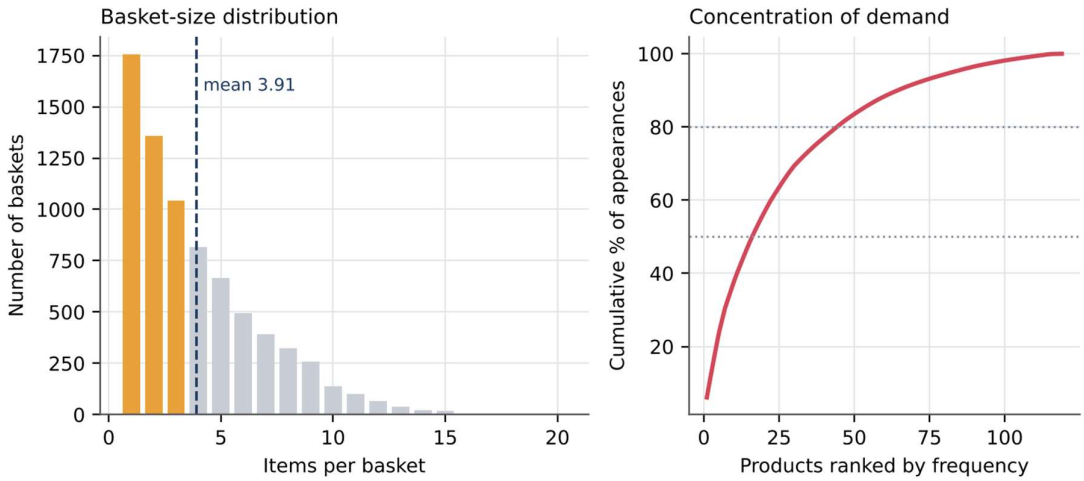


Figure 2. Left: number of baskets by basket size (orange bars mark sizes of three items or fewer). Right: cumulative share of product appearances when products are ranked by frequency.

Single-item baskets are the tallest bar, and the counts then decline steadily to one basket each at 18 and 20 items. The visit pattern is dominated by short trips, so promotions that need a large basket (multi-buy discounts, for example) can reach only a minority, whereas quick-checkout capacity and impulse displays at the till serve the majority.

1.8 Product pairs that occur together

Before clustering, co-occurrence was measured directly. For each pair of products that appear together in at least 0.8% of baskets (224 pairs), the analysis computed support (share of baskets containing both), confidence (probability of the second item given the first) and lift (observed co-occurrence divided by the co-occurrence expected if the two were independent; a value above 1 indicates a positive association).

Pair	Baskets	Support	Lift	Comment
mineral water + spaghetti	448	6.0%	1.44	Most frequent pair
chocolate + mineral water	395	5.3%	1.35	
eggs + mineral water	382	5.1%	1.19	
milk + mineral water	360	4.8%	1.55	
ground beef + herb & pepper	120	1.6%	3.29	Among the highest lifts
olive oil + soup	67	0.9%	2.68	Among the highest lifts
milk + whole wheat pasta	74	1.0%	2.58	
frozen vegetables + tomatoes	121	1.6%	2.47	

The most frequent pairs are driven mostly by the popularity of mineral water, so their lift is modest (1.2 to 1.8). The highest lifts belong to less common pairs such as ground beef with herb & pepper (lift 3.3) and olive oil with soup (2.7), which are the better candidates for a bundle or adjacent shelving. Lift is a screening statistic here; it carries no significance test and the underlying counts are small for some pairs.

1.9 Feature engineering and PCA

Each basket becomes a vector of 119 zeros and ones. PCA was first fitted to all 7,501 baskets to see how variance is spread. The first two components hold only **11.18%** of the variance (PC1 6.72%, PC2 4.47%); 18 components are needed for 50%, 46 for 80% and 66 for 90%. The variance is spread across many weak directions because the products are largely independent of one another. A two-dimensional plot is therefore a visual aid only.

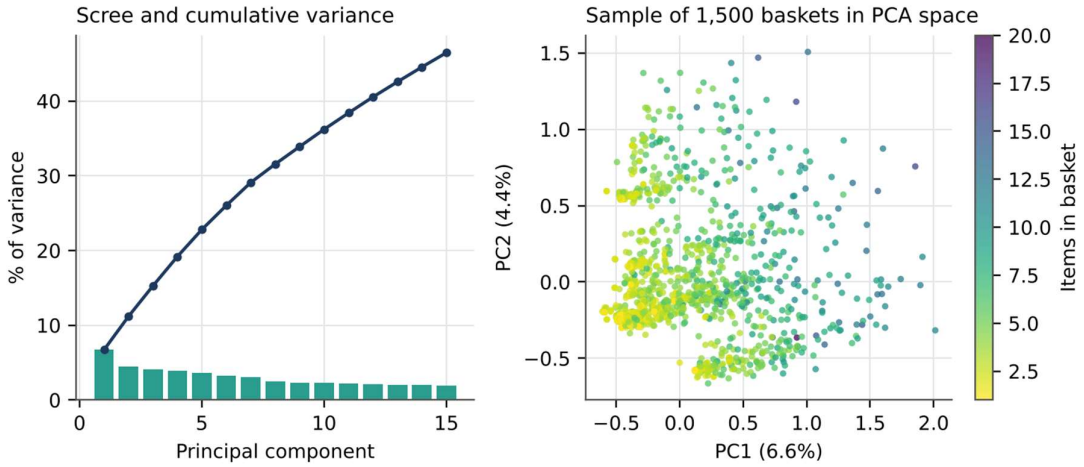


Figure 3. Left: variance explained by the first fifteen components. Right: a random sample of 1,500 baskets projected on the first two components, coloured by basket size.

The projection shows a fan-shaped cloud whose horizontal axis largely tracks basket size: small baskets pile up at the left and large baskets spread to the right. The dense vertical streaks at the left edge are baskets built around a single popular item, and they stand out only because many baskets are identical.

1.10 Clustering methods and evaluation

All algorithms were run on the same random sample of 1,500 baskets (seed 7) so that the comparison is fair. The **silhouette coefficient** ranges from -1 to +1 (higher means better-separated groups; values near zero mean overlapping groups). The **Davies-Bouldin index** is lower for compact, well-separated clusters.

Algorithm	Setting used	Principle
K-means	k = 4, 10 initialisations	Alternates between assigning baskets to the nearest centroid and recomputing centroids
Hierarchical	k = 4, Ward linkage	Merges the pair of groups that increases within-group variance the least
DBSCAN	Jaccard distance, eps = 0.5, min_samples = 5	Grows clusters from dense neighbourhoods; isolated baskets are labelled noise
Spectral	k = 4, 10-nearest-neighbour affinity	Clusters the eigenvectors of a similarity graph

Algorithm	Clusters	Noise baskets	Silhouette	Davies-Bouldin	Cluster sizes
K-means	4	0	0.0693	3.1925	782, 251, 238, 229
Hierarchical	4	0	0.0174	4.7786	832, 330, 193, 145
DBSCAN	2	549 (37%)	0.0872*	1.9344*	945, 6
Spectral	4	0	-0.0246	3.2183	1175, 130, 111, 84

**DBSCAN is scored only on the baskets it did not label as noise, so its numbers are not comparable with the other three rows.*

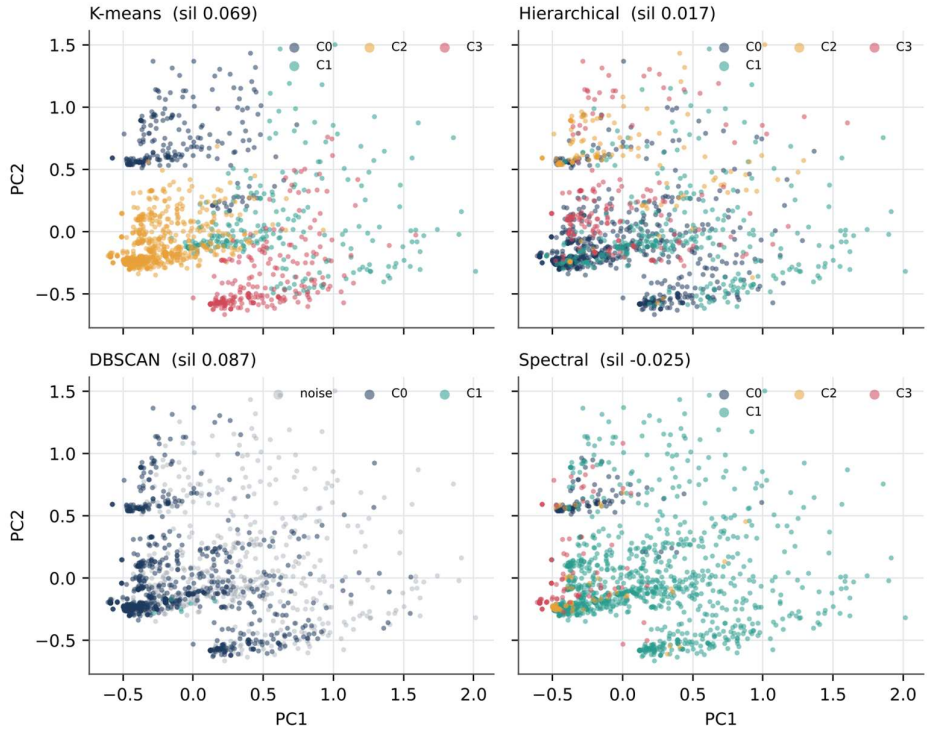


Figure 4. The 1,500-basket sample in PCA space, coloured by the cluster assigned by each algorithm.

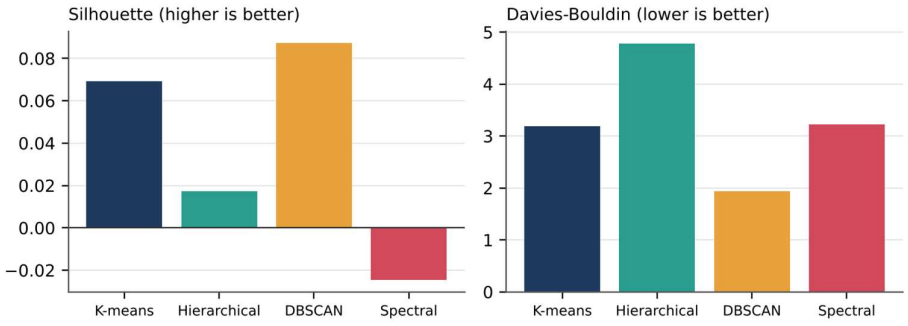


Figure 5. Silhouette (left) and Davies-Bouldin (right) for the four algorithms.

Reading the results

- **K-means** has the best silhouette among the methods that assign every basket (0.069), but the value is close to zero, so the groups overlap heavily. Its clusters are best described by an anchor product: one cluster consists of baskets containing french fries, another spaghetti, a third mineral water (each present in 100% of that cluster's baskets), and the largest cluster (782 baskets, 2.7 items on average) is the remainder of small, mixed baskets. The method splits on the most popular items rather than on genuine shopping missions.
- **Hierarchical clustering** scores 0.017 and the worst Davies-Bouldin value (4.78); its clusters are not distinguishable.
- **Spectral clustering** has a negative silhouette (-0.025) and puts 1,175 of the 1,500 baskets in one cluster, leaving three small groups of 84 to 130 baskets, which is a sign that the similarity graph has no strong community structure.
- **DBSCAN** reports the best-looking pair of scores (silhouette 0.087, Davies-Bouldin 1.93), but it labels 549 baskets (37%) as noise, and its two clusters contain 945 and only 6 baskets, so in practice it finds one dense blob. The score describes a favourable subset, not the data.

Sensitivity checks

Check	Result	What it shows
K-means, k from 2 to 8	Silhouette is highest at k = 2 (0.124) and falls to 0.007 at k = 4 and 0.017 at k = 5. Davies-Bouldin keeps falling from 3.54 at k = 2 to 2.99 at k = 8.	The two internal indices disagree about the best k, and neither value is high.
K-means and Ward on 5 other samples	Mean silhouette 0.080 +/- 0.012 (K-means) and 0.022 +/- 0.021 (Ward)	The weak result is stable across samples; K-means is consistently above Ward.
DBSCAN, eps 0.5 to 0.9 and min_samples 5 or 10	eps = 0.5 gives 2 clusters (549 noise); eps = 0.6 gives 2 clusters with silhouette -0.044; eps of 0.7 or more merges everything into one cluster	Results depend entirely on eps, with no stable intermediate setting.

Taken together, the evidence is consistent: baskets average about four items drawn from 119 products, so two baskets rarely share more than one product, and distance-based clustering has little structure to recover. Silhouette values below 0.1 do not support describing the baskets as belonging to distinct customer segments.

1.11 Business implications

- Use frequency counts to protect availability of the 17 products that generate half of all appearances, and review the long tail separately.
- Design for the small basket: single-item baskets are the most common size, so checkout speed and till-point displays matter more than large-basket offers.
- Test bundles built on high-lift pairs (for example ground beef with herb & pepper) with a small controlled trial before any rollout; popularity-driven pairs such as mineral water with spaghetti already sell together without help.
- Do not build customer segments or targeted campaigns from the cluster labels; the clusters reflect which popular item a basket contains.
- Treat DBSCAN noise as an algorithmic output to inspect, not as proof that a transaction is abnormal.

1.12 Limitations

The file contains no customer identifier, date, price, quantity or product category, so it cannot support statements about customer value, seasonality, revenue or causes of purchase. Binary vectors ignore quantities; PCA retains about 11% of the variance in two dimensions; and all clustering results depend on the algorithm, its parameters and the 1,500-basket sample (although the stability check suggests that the conclusion is not sample-specific).

1.13 Conclusion of Case Study 1

The retail data give a clear descriptive picture: a small group of staples dominates, baskets are short and right-skewed, and a few less common pairs show real positive association. Clustering did not reveal well-separated basket groups under any of the four algorithms or any parameter setting tried. Stronger business conclusions require prices, quantities, dates and customer identifiers.

CASE STUDY 4 - STUDENT DROPOUT AND ACADEMIC SUCCESS

4.1 Problem statement

A higher-education institution wants to know which factors are associated with students dropping out, remaining enrolled or graduating, and to build a baseline model that could help staff offer support early. The target is the student's status at the end of the normal course duration: **Dropout**, **Enrolled** or **Graduate**.

> **Timing note:** because the outcome is recorded at the end of the course, a model meant for early warning may use only information that exists at the time of prediction. All six variables describing the second semester were therefore excluded from the model. First-semester variables were kept as plausible early signals, subject to confirming when they are actually recorded.

4.2 Objectives

- Load and inspect the data, identify the target and classify the predictors.
- Check missing values, duplicates and unusual values.
- Summarise the outcome classes and key numeric variables with suitable charts.
- Compare admission grade, first-semester grade and age across outcomes.
- Test whether first-semester grades differ across outcomes at the 5% level and quantify the size of the difference.
- Train and evaluate a multiclass logistic-regression classifier using per-class and macro-averaged metrics.
- Interpret the model, state limitations and recommend responsible uses.

4.3 Data understanding and quality audit

The file is semicolon-separated, uses a byte-order mark at the start and contains 37 columns: 36 predictors and the Target column. One column header carries a trailing tab character ('Daytime/evening attendance'), which was stripped, and the source spelling 'Nacionality' was kept so that column names still match the supplied file.

Audit item	Result	Comment
Student records	4,424	One row per student
Columns	37	29 integer, 7 decimal and 1 text column (Target)
Missing values	0	No imputation was required, although the pipeline still contains imputers as a safeguard
Exact duplicate rows	0	None
Target classes	3	Dropout, Enrolled, Graduate
First-semester grade equal to 0	718 (16.2%)	Valid in context, since students with no graded unit receive 0, but it forms a spike that distorts means
Zero units enrolled in first semester	180	Likely students who did not take any course in the term; retained
Units enrolled but none evaluated	169	Consistent with absence or deferred assessment; retained
Age at enrolment	17 to 70 years	Long right tail of mature entrants; genuine, not an error

First five records (selected columns)

Marital status	App. mode	Course	Adm. grade	Age	1st sem approved	1st sem grade	Target
1	17	171	127.30	20	0	0.00	Dropout
1	15	9254	142.50	19	6	14.00	Graduate
1	1	9070	124.80	19	0	0.00	Dropout
1	17	9773	119.60	20	6	13.43	Graduate
2	39	8014	141.50	45	5	12.33	Graduate

4.4 Variable families and preprocessing

Many integer columns are really **category codes** (course, application mode, parents' qualification and occupation, nationality). Treating them as quantities would imply an order that does not exist, so they were one-hot encoded. Variables with true numeric meaning were standardised so that large-range variables, such as grades, do not dominate the model.

Family	Variables	Treatment
Target	Target (Dropout, Enrolled, Graduate)	Class label only
Categorical codes	Marital status (6 codes), application mode (18), course (17), previous qualification (17), nationality (21), mother's / father's qualification (29 / 34), mother's / father's occupation (32 / 46), attendance	Most-frequent imputation, one-hot encoding with unseen categories ignored
Binary indicators	Displaced, educational special needs, debtor, tuition fees up to date, gender, scholarship holder, international	Verified to contain only 0 and 1; passed through
Numerical	Age, admission grade, previous-qualification grade, application order, first-semester unit counts and grade, unemployment rate, inflation rate, GDP	Median imputation and standardisation
Excluded	Six second-semester variables	Removed to avoid temporal leakage

The 30 retained predictors expand to 231 model features after encoding. The data were split with a stratified 80/20 division, and every transformation was fitted on the training portion only, inside a single scikit-learn pipeline, so that no test information leaks into training.

4.5 Distribution of outcomes

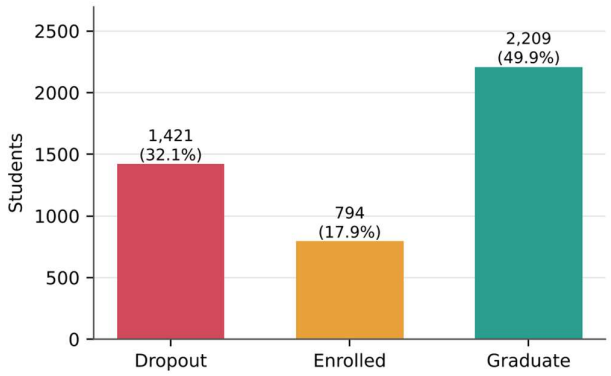


Figure 6. Number and percentage of students in each outcome class.

Outcome	Students	Share
Dropout	1,421	32.1%
Enrolled	794	17.9%
Graduate	2,209	49.9%
Total	**4,424**	**100.0%**

Graduates make up about half of the records, dropouts about a third and Enrolled students under a fifth. The imbalance means that a model predicting 'Graduate' for everyone would already score 49.9% accuracy, so accuracy alone cannot be used to judge the classifier; per-class precision, recall and F1 are required.

4.6 Descriptive statistics and distributions of key variables

Variable	Mean	Std. dev.	Min	Q1	Median	Q3	Max	Skew
Age at enrolment (yrs)	23.27	7.59	17.00	19.00	20.00	25.00	70.00	2.05
Admission grade	126.98	14.48	95.00	117.90	126.10	134.80	190.00	0.53
1st-semester grade	10.64	4.84	0.00	11.00	12.29	13.40	18.88	-1.57

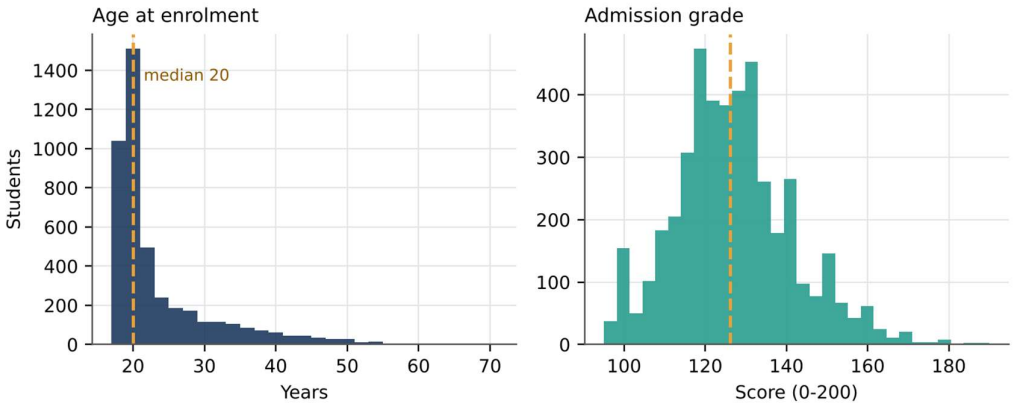


Figure 7. Histograms of age at enrolment (left) and admission grade (right); dashed lines mark the median.

Age is strongly right-skewed (skewness 2.05): most students enrol at 18 to 20, with a tail of mature entrants up to 70. Admission grade is close to bell-shaped and mildly right-skewed, centred near 126 on a 0-200 scale. First-semester grade is left-skewed (skewness -1.57), because the 718 zero grades pull the mean (10.64) well below the median (12.29).

4.7 Admission grade by outcome

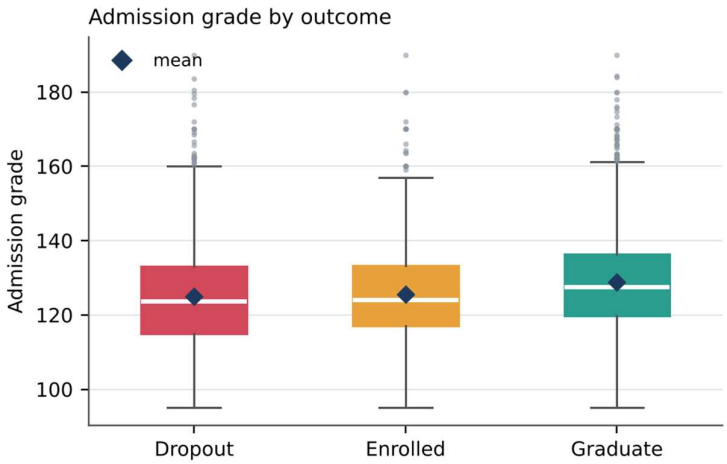


Figure 8. Admission grade by outcome (box plot; diamonds mark the group means).

Outcome	Mean	Median	Std. dev.	Q1	Q3
Dropout	124.96	123.60	15.13	115.00	133.00
Enrolled	125.53	124.10	13.79	117.00	133.28
Graduate	128.79	127.40	14.07	119.70	136.40

Graduates have the highest admission grades (median 127.4 against 123.6 for dropouts and 124.1 for Enrolled students), a gap of about 4 points on a 200-point scale that is small next to the spread of roughly 14 to 15 points within each group. The boxes overlap heavily, so admission grade shifts the typical outcome slightly but cannot by itself predict a student's path.

4.8 First-semester academic performance

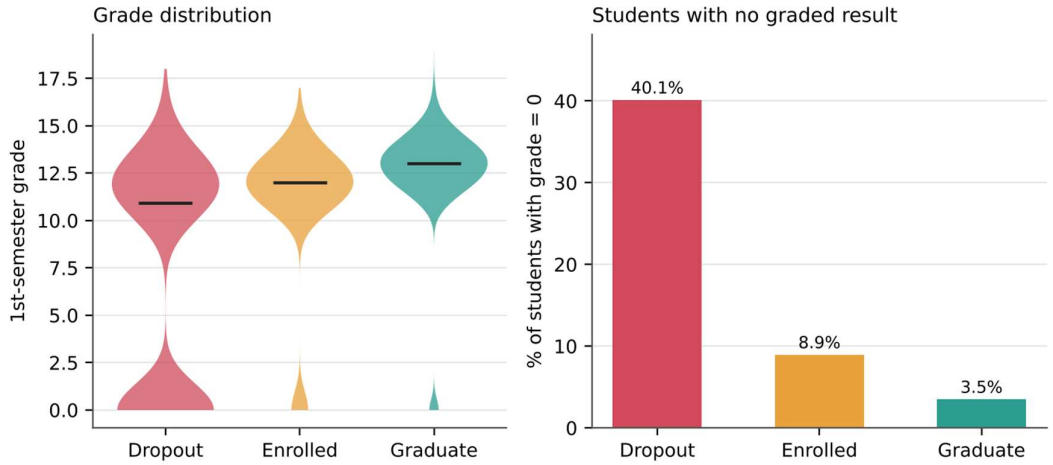


Figure 9. Left: distribution of first-semester grade by outcome (violin plot, black line = median). Right: share of each outcome group with a first-semester grade of zero.

Outcome	Mean grade	Median	Std. dev.	Grade = 0 (count)	Grade = 0 (share)
Dropout	7.26	10.93	6.03	570	40.1%
Enrolled	11.13	12.00	3.68	71	8.9%
Graduate	12.64	13.00	2.70	77	3.5%

This is the sharpest contrast in the data. Among dropouts, **40.1% (570) have a first-semester grade of zero**, compared with 8.9% of Enrolled students and 3.5% of graduates. Dropouts also have the lowest mean grade (7.26) and by far the widest spread (standard deviation 6.03); the median (10.93) sits near that of the other groups because the zeros form a separate sub-population. Among students with a graded first semester, differences are far smaller. Failing to complete any graded unit in the first term is therefore strongly associated with leaving, although the data cannot say why.

4.9 Age at enrolment by outcome

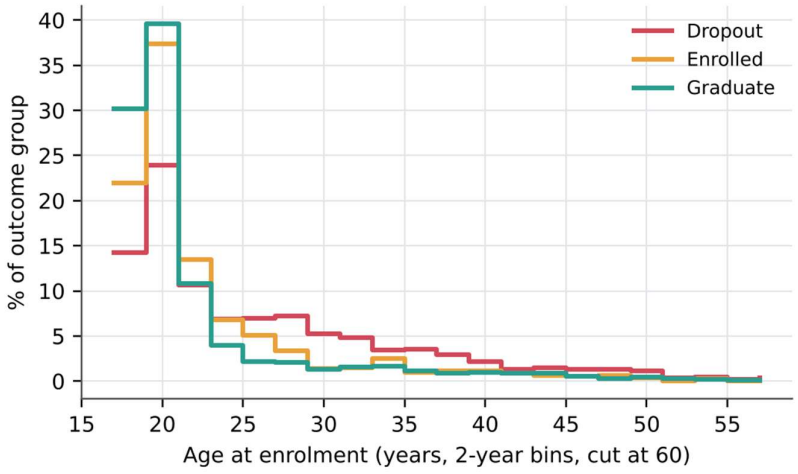


Figure 10. Age at enrolment as a percentage of each outcome group (2-year bins, ages 17 to 60).

Outcome	Mean age	Median age	Std. dev.	Aged over 30
Dropout	26.07	23	8.70	25.0%
Enrolled	22.37	20	6.30	10.6%
Graduate	21.78	19	6.69	10.1%

Dropouts are older on average (median 23 years against 19 for graduates) and a quarter of them (25.0%) enrolled after age 30, compared with about one in ten Enrolled students and graduates. This is a description of who leaves, not an explanation: mature students more often combine study with work or family duties, and age must never be used to label an individual student as high-risk.

4.10 Hypothesis testing

Research question: is the distribution of first-semester grades the same for Dropout, Enrolled and Graduate students? A **Kruskal-Wallis H test** was chosen because it compares three independent groups without assuming normality, and the grade distributions are clearly skewed with a spike at zero (a one-way ANOVA would be inappropriate).

Element	Specification / result
Null hypothesis (H0)	First-semester grade has the same distribution in all three outcome groups
Alternative (H1)	At least one outcome group has a different distribution
Test, level	Kruskal-Wallis H test, alpha = 0.05, 2 degrees of freedom
H statistic	1,094.186
p-value	2.51e-238 (effectively zero)
Decision	Reject H0, because p < 0.05
Effect size	epsilon-squared = H / (n - 1) = 0.247, a moderate-to-large effect

Pair (Mann-Whitney U)	Bonferroni-adjusted p	Rank-biserial r	Reading
Dropout vs Enrolled	2.1e-42	0.35	Dropout ranks lower
Dropout vs Graduate	< 1e-200	0.62	Dropout ranks lower
Enrolled vs Graduate	5.6e-63	0.40	Enrolled ranks lower

The pairwise tests (three comparisons, p-values multiplied by 3) show that all three pairs differ, and the Dropout-Graduate gap is the largest (r = 0.62). For comparison, the same test on **admission grade** gives H = 73.23, p = 1.25e-16, epsilon-squared = 0.017, and on **age** H = 375.11, p = 3.52e-82, epsilon-squared = 0.085. All three are statistically significant, but the practical effect is large for first-semester grade, small-to-moderate for age and very small for admission grade, which illustrates why a p-value alone is not enough: with 4,424 students even a tiny difference is significant.

Assumption	Assessment
Independent observations	Assumed from the file layout (one row per student); this cannot be verified without knowing how records were collected
Independent groups	Each student has exactly one outcome, so the three groups do not overlap
Ordinal or continuous measure	Grades are on a comparable numeric scale
Similar distribution shape	Not met: Dropout is bimodal (zero spike), and Levene's test confirms unequal spread (p < 0.001). The result is therefore read as a difference in distributions, not strictly in medians

4.11 Classification model

A multiclass logistic regression (softmax) was used as a transparent baseline: it trains quickly, produces probabilities and has coefficients that can be inspected. Hyper-parameters were left at their defaults (L2 penalty, C = 1) because the aim is a baseline, not a tuned model.

Detail	Setting
Split	80% train (3,539) / 20% test (885), stratified by outcome, random_state = 2026
Features in / after encoding	30 predictors / 231 model features
Categorical codes	Most-frequent imputation, one-hot encoding
Numerical and binary	Median imputation, standardisation
Classifier	LogisticRegression (multinomial), max_iter = 3000
Excluded	The six second-semester variables
Extra checks	5-fold stratified cross-validation; a class-balanced variant; a retrospective variant that includes second-semester data

4.12 Model evaluation

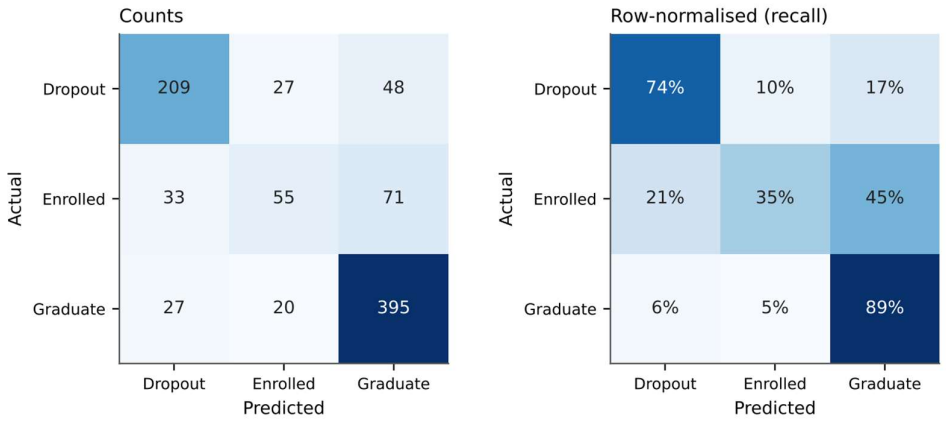


Figure 11. Confusion matrix on the held-out test set: counts (left) and share of each actual class (right).

Actual \ Predicted	Dropout	Enrolled	Graduate	Row total
Dropout	209	27	48	284
Enrolled	33	55	71	159
Graduate	27	20	395	442
Column total	**269**	**102**	**514**	**885**

Metric	Dropout	Enrolled	Graduate	Macro average
Precision	0.777	0.539	0.768	0.695
Recall	0.736	0.346	0.894	0.658
F1-score	0.756	0.421	0.826	0.668
Support	284	159	442	885

Overall metric	Held-out result
Accuracy	0.745 (74.5%)
Macro F1	0.668
Weighted F1	0.731
Majority-class baseline accuracy	0.499
5-fold CV accuracy	0.751 +/- 0.020
5-fold CV macro F1	0.665 +/- 0.026

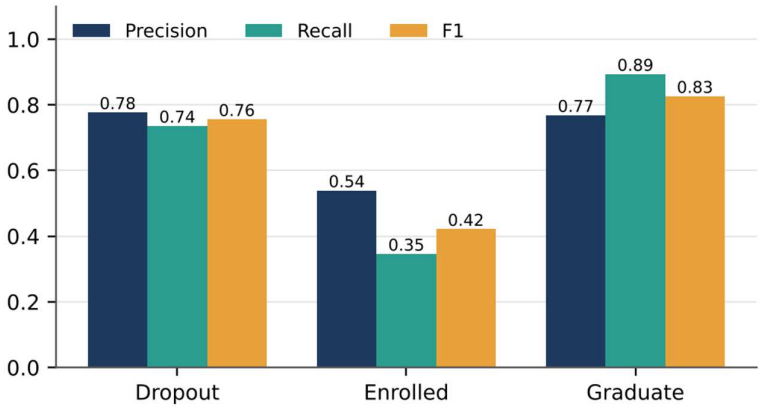


Figure 12. Precision, recall and F1-score for each class on the test set.

Interpretation

- **Graduate** is predicted best (recall 0.89, F1 0.83): 395 of 442 graduates are found. This is the largest class and also the easiest to recognise from first-semester results.
- **Dropout** is identified reasonably well (precision 0.78, recall 0.74): about three in four dropouts are flagged and about the same fraction of dropout flags are correct. 48 dropouts are mistaken for graduates, which is the costly error in a support setting.
- **Enrolled** is the weak class (recall 0.35, F1 0.42): only 55 of 159 Enrolled students are recognised and 71 are labelled Graduate. These students sit between the other two outcomes and are the smallest group, so the model has little to separate them on.
- The accuracy of 74.5% is 25 points above the 49.9% baseline, a real but modest gain, and it hides the poor Enrolled result; macro F1 (0.668) is the more honest summary. Cross-validation (0.751 accuracy) agrees with the single split, so the test result is not an accident of one particular partition.

Effect of class imbalance and of the prediction time

Variant	Accuracy	Macro F1	Dropout recall	Enrolled recall	Enrolled precision
Baseline (reported above)	0.745	0.668	0.736	0.346	0.539
Class-balanced weights	0.714	0.678	0.690	0.604	0.419
Retrospective (adds 2nd-semester data)	0.775	0.708	0.750	0.421	-

Re-weighting the classes raises Enrolled recall from 0.35 to 0.60 but lowers accuracy by about three points and reduces Enrolled precision, so the model simply flags more students and wrongly flags more of them. Whether that trade is worthwhile depends on the cost of an unnecessary outreach compared with a missed student. Adding second-semester data improves macro F1 to 0.708, which shows how much of the later signal is lost when predictions must be made early, yet Enrolled recall stays near 0.42 even then.

4.13 Interpreting the model

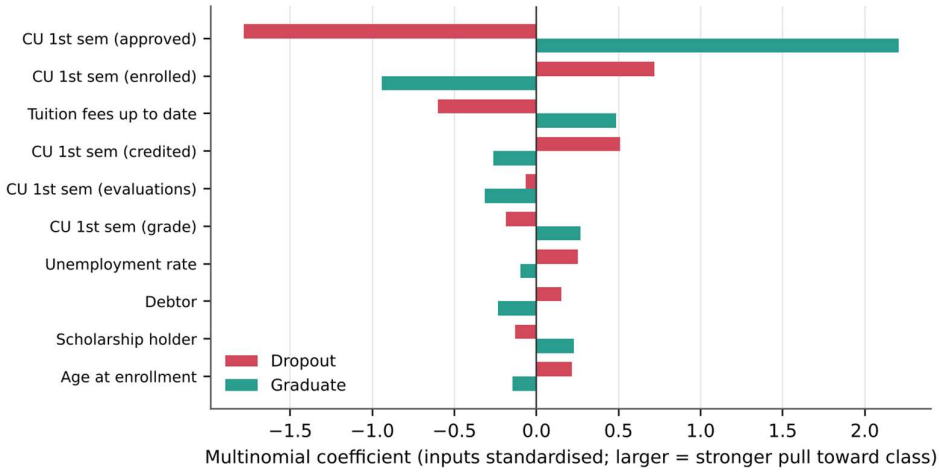


Figure 13. Largest numeric and binary coefficients for the Dropout and Graduate classes (inputs standardised).

Because numeric inputs were standardised, their coefficients can be compared. In the Dropout class the dominant term is the number of first-semester units **approved** (coefficient -1.78, with the opposite sign, 2.21, for Graduate). Students who pass more units are less likely to be predicted as dropouts, and holding approved units constant, a higher count of units enrolled raises dropout odds (0.72), which describes students who register for many units but pass few. Other terms point the same way: tuition fees up to date (-0.60 for Dropout, 0.49 for Graduate), being a scholarship holder (-0.13 for Dropout), being a debtor (0.15), higher age at enrolment (0.22) and a higher unemployment rate (0.25). Admission grade has a near-zero coefficient (-0.05) once first-semester results are known, which matches the weak univariate result in Section 4.7.

Three cautions apply. First, the approved, enrolled, evaluated and credited unit counts are correlated, so individual coefficients may trade off against one another; they are best read together. Second, the largest coefficients overall belong to one-hot categories with few students (for example particular parental occupation or qualification codes), and these are unstable and cannot be labelled without the code book, so they were not interpreted. Third, none of these coefficients shows that a characteristic causes dropout: financial status, for example, may reflect circumstances that also affect study time.

4.14 Limitations of the student analysis

Limitation	Why it matters
Class imbalance	Enrolled is hardest to identify (recall 0.35); accuracy overstates quality
Missing context	Health, family, work, motivation and institutional support are not recorded
Single institution and period	Performance on another institution or later cohort is unknown; the macro-economic columns repeat across students within a period
Linear baseline	Logistic regression cannot capture interactions; tree-based or boosted models might perform differently, although they are less transparent
Prediction time	Every predictor must be confirmed as available when the prediction is made
Association, not causation	Coefficients and group differences do not show why students leave
Fairness and privacy	Age, gender, nationality and family background raise discrimination and data-protection concerns; the model's error rates were not audited by subgroup

4.15 Evidence-based recommendations

Recommendation	Evidence	Responsible action
Offer early academic support	40.1% of dropouts had no graded first-semester result; units approved is the strongest model term	Contact students with no passed units early in the first term with voluntary tutoring, mentoring and advising
Check financial pressure	Fees up to date, debtor and scholarship status carry consistent coefficients	Offer fee counselling and flexible payment options to students who ask or are referred, without making this a ground for penalty
Support mature entrants	One quarter of dropouts enrolled after age 30	Provide evening or flexible schedules and part-time advice; never use age alone as a risk label
Evaluate every class	Enrolled recall is only 0.35	Monitor per-class recall and macro F1 each term; do not optimise accuracy alone
Keep human oversight and validate	False negatives and false positives both occur; one split and one cohort	Advisers review each flag with the student; re-test on later cohorts and audit error rates by subgroup before and during use

4.16 Conclusion of Case Study 4

First-semester performance is strongly associated with the final outcome (Kruskal-Wallis H = 1,094.19, $p < 0.05$, epsilon-squared about 0.25), with no graded result in the first term being the most striking marker of later dropout. Admission grade and age differ by outcome but with small and overlapping effects. A baseline logistic regression using only first-semester information reaches 74.5% accuracy and macro F1 of 0.668, detecting about three in four dropouts but only about one in three Enrolled students. It is suitable as a prompt for supportive conversations, not as an automatic decision-maker.

DISCUSSION AND CLOSING

Cross-case conclusion

The retail study shows that simple counts and co-occurrence measures are informative while clustering of sparse, short baskets is not, and it demonstrates why internal validity scores must be read with the number of points they cover. The student study shows how class imbalance, prediction timing and effect size change the meaning of an apparently good result. In both cases the choice of method has to fit the structure of the data, and the conclusions have to stay within the evidence.

Retail data (Case Study 1)	Student data (Case Study 4)
No customer ID, date, price or quantity	Family, health and institutional factors not recorded
PCA captures only 11.2% of the variance in two dimensions	Enrolled is the smallest class and the hardest to detect
Cluster results depend on algorithm, parameters and sample	Predictors must be available at the time of prediction
Lift identifies association, not purchase motives	Prediction and association do not prove cause

Reproducibility checklist

Item	Action
Data	Keep the supplied files unchanged. Read the retail CSV with header=None and the student CSV with sep=';' and encoding='utf-8-sig'; strip column names.
Preparation	Strip and lower-case product labels, drop blank cells and repeated items in a basket; drop the six second-semester columns before modelling.
Seeds	Clustering sample: seed 7 (1,500 baskets). Train/test split and cross-validation: seed 2026. K-means and spectral clustering use the same sampling seed.
Model	Stratified 80/20 split; pipeline = imputation + one-hot / scaling + multinomial logistic regression (max_iter = 3000).
Outputs	All figures, tables and metric values in this report come from one scripted run; re-running the notebook with the same seeds reproduces them.
Ethics	Use model outputs to support human review and student assistance, never for automatic penalties.

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